

- 1 Doctors use tests to see if there are changes in a person's body compared to normal. List three tests a doctor may perform.

i

ii

iii

- 2 A doctor tells a patient that they have acne. Suggest what evidence the doctor has found to make them think this.

- 3 What does 'evidence' mean? Tick (✓) *one* box.

- ☐ A the name given to a certain type of disease
☐ B changes in the way that the body works
☐ C information used to say whether something is right or wrong
☐ D a type of test performed by a doctor

- 4a Which of the following best describes the heart? Tick (✓) *one* box.

- ☐ A an organ
☐ B a cell
☐ C a tissue
☐ D a system

- b State one job that the heart does. _____

- 5 In groups, discuss how to tell if something is alive. Record your group's ideas in the first box below. Ignore the other box for the moment! There is no right or wrong answer.

Ideas from my group:

What we now think:

1 Tick (✓) the living organisms in this list.

☐ elephant

☐ soccer ball

☐ robot

☐ pine tree

☐ aeroplane

☐ human

☐ desert

☐ mouse

2 A mnemonic (pronounced 'nem-on-ic') is a word or phrase that helps you to remember a list. MRS GREN is a mnemonic to remember the life processes.

a What is a life process? _____

b Draw *one* line from each letter to a life process. Then draw *one* line from each life process to its meaning. One has been done for you.

	Life process	Meaning
M	nutrition	increasing in size
R	respiration	getting rid of waste
S	reproduction	making new life
G	growth	releasing energy
R	movement	changing position
E	sensitivity	detecting changes in the surroundings
N	excretion	needing substances to carry on living

3 What do humans need to respire? Tick (✓) *two* boxes.

☐ oxygen

☐ carbon dioxide

☐ wood

☐ fire

☐ food

☐ light

4 Describe one way in which the growth of trees is different from the growth of humans.

5a In what ways is a car like an organism? _____

b Why is a car not an organism? _____

6 In your group, look at your answers to question 5 on page 3. Discuss your ideas again to find out if they have changed. If you can, write a better answer in the lower box.

1 Use a *pencil* to tick (✓) *one* box to answer to each question.

a Which organ lets you think?

- ☐ **A** heart
- ☐ **B** brain
- ☐ **C** skull
- ☐ **D** liver

b Which organ pumps blood?

- ☐ **A** heart
- ☐ **B** kidney
- ☐ **C** lungs
- ☐ **D** liver

c A job of the stomach is to:

- ☐ **A** store waste materials.
- ☐ **B** make urine.
- ☐ **C** break up food.
- ☐ **D** help you breathe.

d A job of the liver is to:

- ☐ **A** destroy some substances.
- ☐ **B** absorb food.
- ☐ **C** make urine.
- ☐ **D** excrete water.

e A sheep's lungs excrete:

- ☐ **A** mucus.
- ☐ **B** carbon dioxide.
- ☐ **C** urine.
- ☐ **D** oxygen.

f The tube taking food to the stomach is the:

- ☐ **A** intestine.
- ☐ **B** diaphragm.
- ☐ **C** oesophagus.
- ☐ **D** trachea.

2 Ask your teacher how many of your answers to question **1** are correct. Then work with others and the Student Book to find your incorrect answers. Make corrections and check with your teacher. Do this until they are all correct.

3 Give the name of the organ that humans use for sensitivity. _____

4 Give the name of the organ that plants use for nutrition. _____

5 State two functions of a plant root.

i _____

ii _____

6a What process produces the food stored in plant storage organs?

b Why won't a potato grow if the potato plant does not get much light?

1 Complete the sentences using words from the box. Use each word once.

conclusion diagnosis doctors evidence explain test

Scientists invent ideas to try to _____ things that they see happening. They then do experiments to _____ their ideas. They use the results from their tests as _____, to decide if an idea is correct or incorrect. This decision is called a _____. Medical _____ also use this process to try to work out what is wrong with someone. A conclusion about what is wrong with someone is called a _____.

2 Look at the drawing.

a What organ is the doctor examining?

b What is the function of this organ?



3 Draw *one* line from each symptom to the organ that is the cause.

Evidence (or 'symptom')

only small amount of urine produced

difficulty breathing

very fast pulse

sore chest after eating food

Diagnosis

stomach problem

kidney problem

lung problem

heart problem



4 You can use these items to make different stethoscopes.

a Describe how to test a stethoscope. _____

b Explain how to judge which design of stethoscope is best. _____

- 1 Complete the sentences using words from the box.
Use each word more than once.

organ tissue

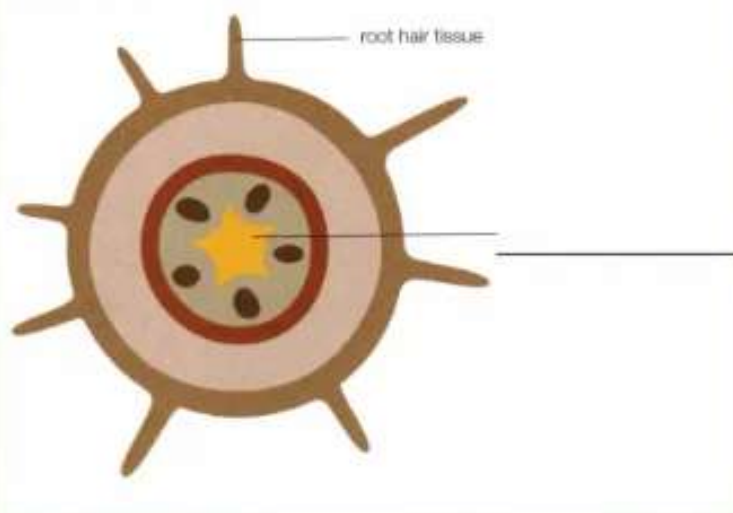
Every _____ in your body is made of many different
types of _____. An example of a _____ is fat.
An example of an important _____ in your body is your liver.

- 2 The diagram shows a slice through a plant root.

a What is the function of root hair tissue?

b On the diagram, write in the name of the central
tissue (which carries water in a plant).

c How many tissues in total are found in this root?



- 3 Explain how you can identify one of the tissues
in your tongue. To answer this, tick (✓) one box
for each of parts **a** and **b** below.

a A tissue in your tongue is:

- ☐ **A** hair.
☐ **B** root hair tissue.
☐ **C** bone.
☐ **D** muscle.

b You know this because:

- ☐ **A** your tongue is hard.
☐ **B** your tongue feels furry.
☐ **C** you can move your tongue.
☐ **D** your tongue is wet.

- 4 The leaf of a plant is an organ.

a Give a function of a leaf. _____

b Suggest one tissue that it contains. _____

c Explain why you think it contains this tissue. _____

- 5 Complete the table.

Organ	Example of tissue	Function of tissue
heart		

- 6 What sort of organ is a carrot? _____

- 1** Describe what a microscope is used for.

- 2** Complete the labels on the microscope, to show the names of the parts.

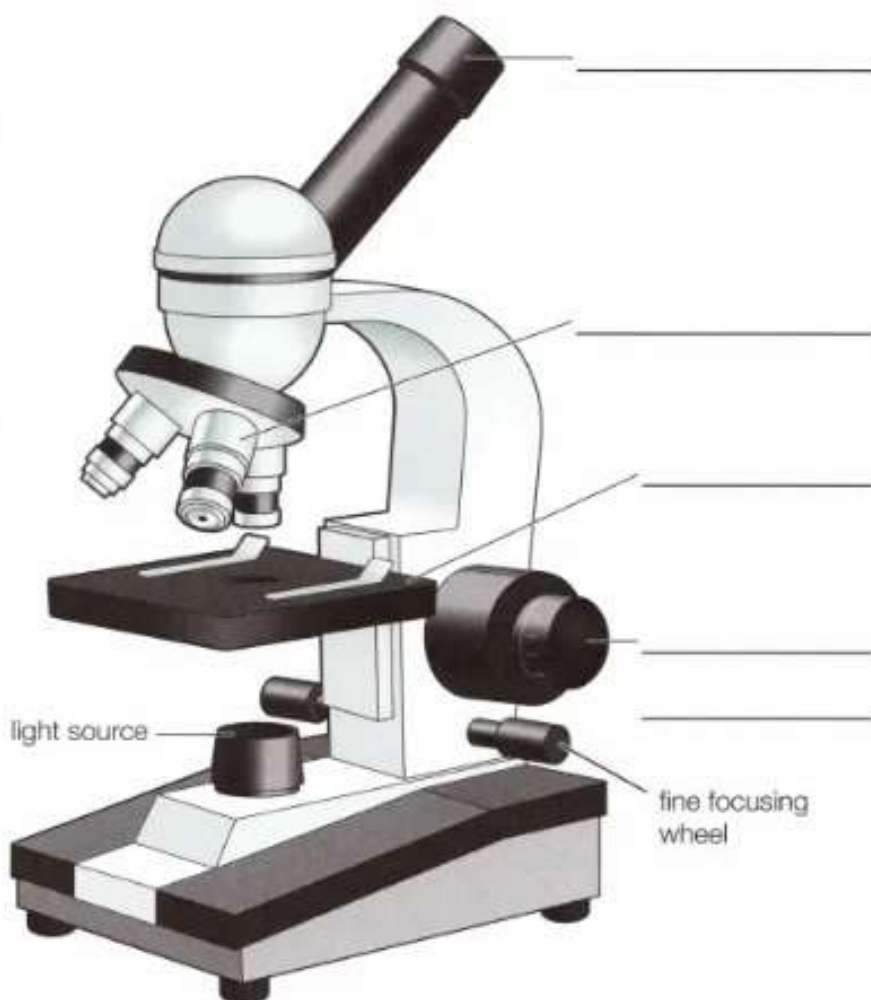
- 3a** Number these sentences in order, to describe how to use a microscope.

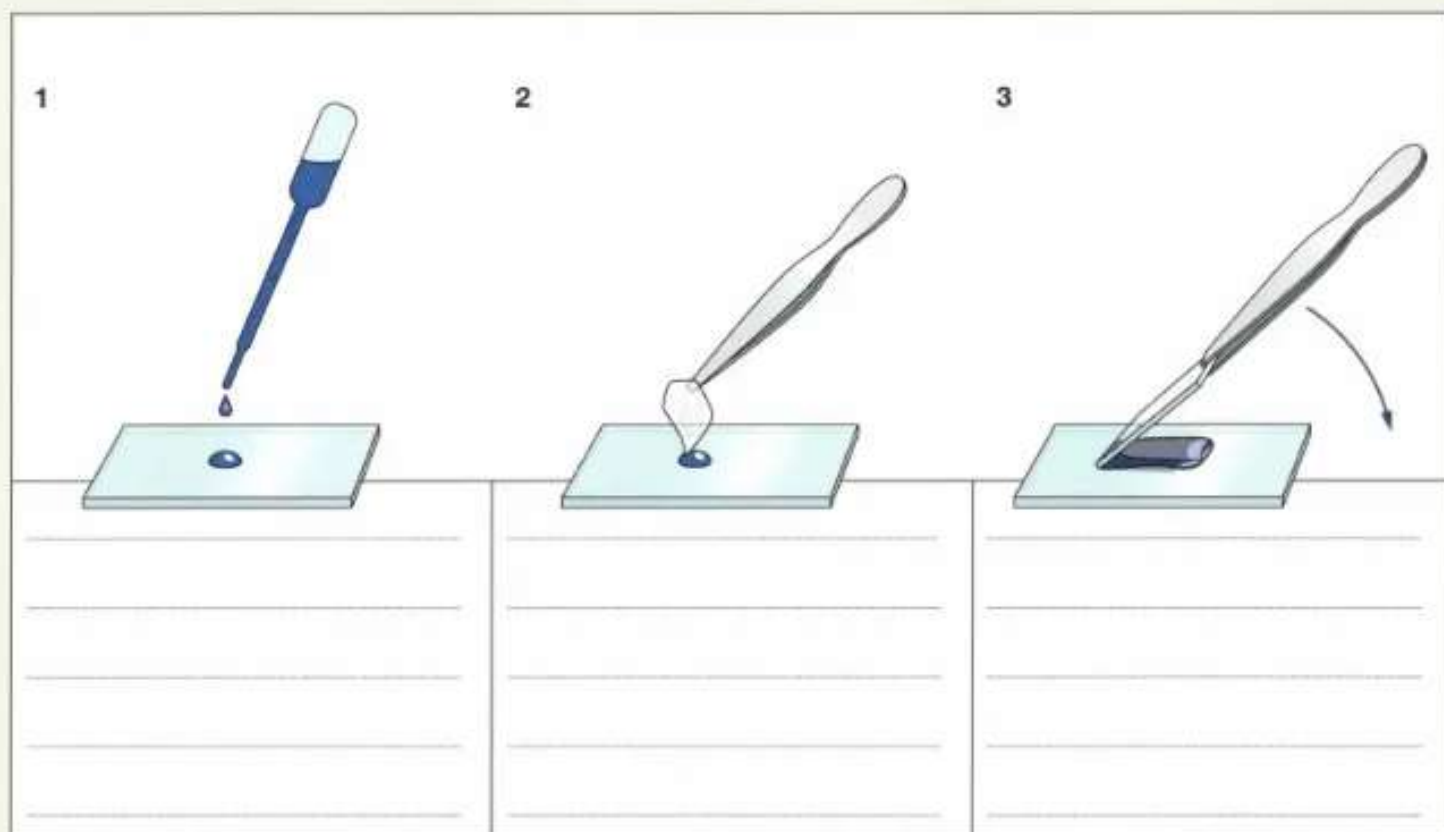
- ☐ Look down a microscope with both eyes open and adjust the light source.
- ☐ Make the gap between the stage and the lens as small as possible.
- ☐ Place a slide on the stage. Use the clips to hold the slide in place.
- ☐ Move the lowest power lens over the hole in the stage.
- ☐ To magnify more, use the next most powerful lens. Use the fine focusing wheel to adjust the focus.
- ☐ Widen the gap between the stage and the lens, until what you see is clear (in focus).

- b** Describe how you would 'widen the gap', between the stage and the lens.

- 4** What is a specimen? _____

- 5a** Write instructions for these diagrams to describe how to make a slide.





b Share your instructions with others. Discuss which instructions are best and why. On the lines below, write an improved version of *one* of your instructions.

c How have you improved this instruction?

6 Why do we use coverslips?

7 Complete the table.

Eye-piece lens power	Objective lens power	Total magnification
$\times 5$	$\times 10$	
$\times 7.5$	$\times 15$	
	$\times 20$	$\times 200$